Global Noise Reduction versus Phonetic Preservation in Wiener Speech Enhancement

Weitao Shi | SID 550603890 | ELEC5305 Project Report

Project site: https://saotao117.github.io/elec5305-project-550603890/

Code: https://github.com/SAOTAO117/elec5305-project-550603890

Contents

Abstract 2
1 Introduction 2
2 Literature Review 2
2.1 Enhancement and speech distortion 2
2.2 Phonetic structure and evaluation 3
2.3 Method selection 3
3 Methods 3
3.1 Data and experimental conditions 3
3.2 Mixture generation and Wiener filtering 4
3.3 Phonetic measurements 5
3.4 Aggregation and validation 5
4 Results 6
4.1 Whole utterance performance 6
4.2 Speech retention and class differences 7
4.3 Gain and spectral preservation 8
4.4 Window length comparison 10
5 Discussion 10
5.1 Interpretation of the main findings 10
5.2 Relation to previous work 10
5.3 Window length and signal preservation 11
5.4 Limitations and further work 11
6 Conclusion 11
Reproducibility 12
References 12

Abstract

This study examines phonetic preservation in speech enhancement using the short-time Fourier transform (STFT) and Wiener filtering. Ten labelled English utterances from ten speakers were mixed with three environmental noises at four signal-to-noise ratios (SNRs), with two excerpts per utterance–noise pair. The 240 mixtures were processed using a 32 ms window. Applying the estimated gain separately to clean speech and noise distinguished speech attenuation from noise suppression. Mean error SNR improved by 6.18 dB, while mean short-time objective intelligibility (STOI) changed from 0.8432 to 0.8385. Speaker-balanced speech retention was −1.24 dB for vowels, −2.53 dB for fricatives and −2.51 dB for stops and affricates. Unweighted gain and high-frequency retention did not follow the same class ranking. A paired 16 ms comparison reduced boundary-crossing frames but increased speech attenuation and lowered STOI. The results establish unequal signal preservation in this sample; they do not establish the perceptual loss of particular speech sounds.

1 Introduction

A Wiener enhancer attenuates spectral regions in which estimated speech power is small relative to noise power. The rule is interpretable, but it does not distinguish a weak consonant from an unwanted noise component. Vowels, fricatives and stops differ in energy, periodicity and duration, so a whole-sentence score can obscure how they are processed. A strong vowel contributes much more to a global energy sum than a short, weak consonant.

The research question is whether STFT Wiener filtering preserves broad phonetic classes unequally, and whether improved global error SNR can coexist with attenuation of desired speech. The investigation uses controlled mixtures and phoneme timing labels to compare gain, speech and noise retention, spectral distortion and segmental error SNR. It tests the direction of the class differences rather than assuming that fricatives must be the most damaged sounds.

Gain-distortion experiments [2] and phoneme-scale evaluations of multichannel systems [3] motivate this study, but do not directly answer the question for the selected single-channel Wiener rule. The contribution is its evaluation at utterance and phone scales, followed by a paired change in window length. The filter itself is a standard method.

2 Literature Review

2.1 Enhancement and speech distortion

Loizou [1] describes the Wiener estimator through a mean-square-error objective and the ratio of speech to total spectral power. This gives a useful baseline because the gain can be inspected directly. Its practical weakness is dependence on estimated powers: changing noise or weak speech can produce a poor local SNR estimate. A fixed gain floor limits attenuation but does not encode phonetic importance. This project therefore uses a single transparent configuration and investigates its behaviour instead of comparing many enhancement systems.

Kim and Loizou [2] separated gain-induced amplification and attenuation distortions in controlled listening experiments. Their attenuation-only conditions had relatively small effects on intelligibility. This is a crucial qualification: an energy decrease cannot by itself demonstrate a linguistic loss. Their work motivates separating the desired component from interference, while also requiring caution when interpreting that decomposition. The present experiment measures component energy and an objective intelligibility score; it does not reproduce their listening experiment or their exact gain rule.

2.2 Phonetic structure and evaluation

Monir, Magron and Serizel [3] assessed three multichannel systems at phoneme scale. They reported category-dependent behaviour, including difficulty with plosives and improvements for some nasals and sibilants. The result supports local evaluation, but not a universal ranking of vulnerable sounds. Their spatial processing, room conditions and evaluation measures differ from this single-channel experiment. The unresolved question here is how an estimated classical Wiener gain interacts with the labelled sounds under the selected additive noises.

Kong, Mullangi and Kokkinakis [4] classified fricatives using short-frame spectral features and examined robustness to noise. Their work shows why spectral shape above 1 kHz matters, rather than treating all frication energy as interchangeable. Shadle and colleagues [5] further demonstrate that fricative measurements depend on analysis choices and that informative structure extends into high frequencies. These studies motivate separate frequency-band measurements. A 16 kHz recording, however, only permits analysis up to 8 kHz; the present high-frequency measure covers 4–8 kHz, not the complete fricative spectrum.

STOI, introduced by Taal and colleagues [6], predicts intelligibility using short-time spectral-envelope information. It is useful as an utterance-level reference, but it does not identify the phone responsible for a change. Error SNR, spectral distortion and speech retention also answer different questions. The evaluation consequently combines these measures without treating any one of them as a direct count of correctly perceived phonemes.

2.3 Method selection

Spectral subtraction is a simple alternative, but residual musical-noise artefacts can remain [1]. Multichannel methods exploit spatial information but require several microphone signals [3]. Wiener filtering provides an explicit gain linked to mean-square-error estimation and suits the available single-channel data. Its dependence on power estimates remains a weakness, especially for weak speech and changing noise.

Controlled mixing provides the exact clean and noise components needed for decomposition. LibriSpeech supplies open audiobook recordings [7]; the selected derivative resource supplies existing phone boundaries [9]. DEMAND supplies recorded environmental noise [8]. This route avoids making a new alignment system part of the project, at the cost of relying on supplied rather than manually reannotated boundaries. MATLAB provides STFT and inverse-STFT operations [10], while mixture generation, Wiener gain estimation and phonetic measurements are implemented in the project code.

Two implementation concepts are central. First, the gain must be held fixed when decomposing speech and noise; re-estimating it on each reference would define a different experiment. Second, assigning a frame by its centre prevents double counting but does not confine the analysis window to that phone. The methods below address the first issue with a numerical decomposition check and investigate the second with a matched window-length comparison.

3 Methods

3.1 Data and experimental conditions

The sample contains the first eligible 16 kHz mono utterance from each of ten different speakers in the dev.clean split of [9]. Eligibility requires 3–8 s duration, all four speech classes and silence, known phone symbols and continuous labels. The 549 segments comprise 192 vowels, 113 nasals and approximants, 104 fricatives, 97 stops and affricates, and 43 silence segments. The fricative

class includes voiced sounds and HH; stop labels cover the segment rather than an isolated burst. Silence is a separate gain control and is excluded from speech-retention comparisons.

The three noise recordings are kitchen noise (DKITCHEN), cafeteria noise (PCAFETER) and traffic noise (STRAFFIC), using their first channel. A fixed random seed selects two excerpts for each utterance–noise pair, and each excerpt is reused at −5, 0, 5 and 10 dB. The design has 240 mixtures per configuration. Excerpts can overlap, so repeats increase condition coverage without increasing the number of independent speakers.

Table 1. Filter settings and controlled window comparison.

Parameter	Baseline	Comparison
Sampling rate	16 kHz	16 kHz
Window	512 samples, 32 ms	256 samples, 16 ms
Analysis window	Square-root periodic Hann	Same window family
Hop and overlap	256 samples, 50%	128 samples, 50%
FFT length	512 points	512 points
Mixture-power smoothing	0.8	0.8944
Noise estimate	0.5 s noise-only prefix	Same prefix
Minimum amplitude gain	0.05	0.05

These settings define a fixed baseline. The 0.8 smoothing factor corresponds to a time constant of about 72 ms at the 16 ms hop; the gain floor limits per-bin attenuation to about 26 dB. The experiment varies window length without claiming that these other settings are optimal.

3.2 Mixture generation and Wiener filtering

Let A be the samples labelled as active speech, s[n] the clean utterance and v[n] the selected noise. Sums indexed by A or I run over the samples in that set. The mixture and noise scale are

$$y[n] = s[n] + \alpha v[n], \quad \alpha = \left[\frac{\sum_A s[n]^2}{\sum_A v[n]^2 \cdot 10^{\mathrm{SNR}/10}}\right]^{1/2} \tag{1}$$

A 0.5 s noise-only prefix precedes each utterance. Input SNR calibration uses A; whole-utterance evaluation excludes the prefix but includes labelled silence. The target SNR can therefore differ from the whole-utterance input error SNR. Denote the scaled noise by n[n] = αv[n].

Let x be the signal after boundary zero padding, w the analysis window, L its length, H the hop and K the transform length. With frequency-bin index k, frame index m, within-window index l and imaginary unit j, the STFT is

$$X[k,m] = \sum_{l=0}^{L-1} x[mH+l]w[l] \exp(-j2\pi kl/K) \tag{2}$$

Y, S and N denote the STFTs of the mixture, clean speech and scaled noise. Frame centres are converted back to utterance coordinates by removing the padding and prefix offsets. Noise power $P_n[k]$ is the mean of $|Y[k,m]|^2$ over windows wholly inside the prefix and is then fixed. Mixture power is initialised from this estimate and smoothed with β = 0.8:

$$P_y[k,m] = \beta P_y[k,m\text{-}1] + (1\text{-}\beta)|Y[k,m]|^2 \tag{3}$$

$$P_s[k,m] = \max(P_y[k,m] - P_n[k], 0) \tag{4}$$

$$G[k,m] = \max(0.05, \frac{P_s[k,m]}{\max(P_s[k,m] + P_n[k], \varepsilon)}) \quad (5)$$

The numerical floor ε is the larger of 10^{-12} times the maximum mixture STFT power and realmin. For uncorrelated speech and noise, minimising $E\{|S-GY|^2\}$ gives the ideal real gain $P_s/(P_s+P_n)$. Equations (3)–(5) approximate this rule with estimated powers and a lower gain bound. The gain is calculated from the noisy input and its noise-only prefix; clean reference speech is not used to estimate it.

The enhanced spectrum is GY, retaining the mixture phase. Inverse STFT (ISTFT) uses weighted overlap-add with the same window, normalising overlapping products by the window-square sum. The square-root Hann window and 50% overlap support unity-gain reconstruction. After removing boundary padding, the enhanced signal ŷ and its speech and noise components are

$$s_G = \mathrm{ISTFT}(GS), \quad n_G = \mathrm{ISTFT}(GN), \quad \hat{y} = s_G + n_G \quad (6)$$

3.3 Phonetic measurements

Each STFT frame is assigned once to the phone containing its centre. For a phone with exact sample interval I, the primary speech and noise retention measurements are

$$R_s(I) = 10 \log_{10} \frac{\sum_I s_G[n]^2}{\sum_I s[n]^2}, \quad R_n(I) = 10 \log_{10} \frac{\sum_I n_G[n]^2}{\sum_I n[n]^2} \quad (7)$$

More negative values indicate stronger attenuation. Both ratios use the same time interval after synthesis. Overlap-add can redistribute energy across a short interval, so occasional positive speech-retention values are retained. A separate frame-energy ratio is used only as a boundary diagnostic. The 4–8 kHz measure uses clean STFT energy and the same gain in centre-assigned frames, with one-sided energy weights of two except at DC and Nyquist.

Unweighted gain averages G across one-sided bins and assigned frames. Energy-weighted gain uses $|S|^2$ and the one-sided factors as weights. It averages amplitude gain, whereas frame-energy retention averages squared gain; neither is identical to exact-interval retention after synthesis. Gain is also averaged below 1 kHz, from 1–4 kHz and from 4–8 kHz.

$$\mathrm{SNR}_e(I,z) = 10 \log_{10} \frac{\sum_I s[n]^2}{\sum_I (z[n]-s[n])^2} \quad (8)$$

The error-SNR change is $\mathrm{SNR}_e(I,\hat{y}) - \mathrm{SNR}_e(I,y)$. Using the full utterance gives the global score; using each phone interval gives the segmental score. Output error combines speech modification and residual noise. Log-spectral distortion (LSD) is calculated as

$$\mathrm{LSD} = \frac{1}{M} \sum_{m=1}^{M} \left[\frac{1}{B} \sum_{k=0}^{B-1} (D_{\text{processed}}[k,m] - D_{\text{clean}}[k,m])^2 \right]^{1/2} \quad (9)$$

M counts assigned frames and B counts one-sided bins. D denotes log magnitude, calculated using 20 $\log_{10}$ after flooring magnitudes at 10^{-3} times the maximum clean STFT magnitude. Mixture LSD compares GY with S; speech-component LSD compares GS with S. STOI uses each full utterance. Speech metrics exclude segments below 10^{-12} of utterance speech energy. High-frequency ratios require band energy above 10^{-10} of total weighted clean STFT energy across the phone's assigned frames. Missing-frame spectral metrics are recorded as NaN, with valid counts retained.

3.4 Aggregation and validation

Measurements are averaged over repeats within each phone, then over phones within each speaker, noise, SNR and class. Class summaries give each speaker equal weight. Pooled results

first average conditions within speaker. Reported energy ratios are means of dB scores, not ratios formed after pooling all samples. Error bars show between-speaker standard deviations. Paired class contrasts subtract each speaker's vowel score in the same condition; percentile intervals resample ten paired differences 10,000 times with seed 5305. These exploratory intervals are not adjusted for multiple comparisons.

The maximum relative reconstruction and decomposition errors are 2.70×10^{-16} and 3.03×10^{-16}. Additional checks cover unity gain, gain limits, independence from clean reference components, one-sided energy weights and missing-frame handling. Active SNR is checked for every mixture. All 549 tokens receive at least one baseline frame, but 40 of the 506 speech tokens have no 32 ms frame wholly within the label.

The paired 16 ms experiment uses identical speech, labels, noise crops and scaling. Setting β to the square root of 0.8 preserves the smoothing time constant when the hop is halved. A 512-point FFT maintains the frequency grid through zero padding, although true frequency resolution changes. Primary comparisons use time-domain retention, global error SNR and STOI, together with the fraction of frames crossing phone boundaries.

4 Results

4.1 Whole utterance performance

Mean global error SNR improves in all twelve noise–SNR conditions (Table 2), with a pooled improvement of 6.18 dB. Averaging the four SNRs gives 7.29 dB for kitchen noise, 4.39 dB for cafeteria noise and 6.86 dB for traffic noise. These differences describe the selected recordings; they do not isolate a causal effect of noise stationarity.

Table 2. Global error-SNR improvement and STOI change. SNR values are mean ± SD across ten speakers after averaging the two noise excerpts.

Noise	Input SNR (dB)	Error-SNR change (dB)	STOI change
Kitchen	-5	7.54 ± 1.94	+0.0060
Kitchen	0	7.42 ± 1.89	+0.0024
Kitchen	5	7.24 ± 1.83	+0.0007
Kitchen	10	6.97 ± 1.78	+0.0001
Cafeteria	-5	5.29 ± 0.58	-0.0389
Cafeteria	0	4.69 ± 0.58	-0.0191
Cafeteria	5	4.09 ± 0.63	-0.0068
Cafeteria	10	3.51 ± 0.68	-0.0003
Traffic	-5	7.88 ± 2.28	-0.0002
Traffic	0	7.18 ± 1.94	-0.0001
Traffic	5	6.49 ± 1.75	-0.0004
Traffic	10	5.90 ± 1.73	+0.0004

Mean STOI decreases from 0.8432 to 0.8385. Cafeteria noise produces the largest decrease: −0.0389 at −5 dB despite a 5.29 dB error-SNR improvement. The gain therefore improves the energy-error score without improving the objective intelligibility score in that condition.

4.2 Speech retention and class differences

Table 3. Pooled phone measurements. Energy ratios are mean ± SD across ten speaker means, with equal weight for noise and SNR conditions.

Class	Speech retention (dB)	Noise retention (dB)	Mean gain	Weighted gain
Vowels	-1.24 ± 0.31	-8.67 ± 1.88	0.450	0.875
Nasals and approximants	-1.48 ± 0.51	-8.84 ± 2.13	0.424	0.851
Fricatives	-2.53 ± 1.14	-9.02 ± 1.90	0.465	0.749
Stops and affricates	-2.51 ± 0.85	-9.01 ± 2.11	0.422	0.742

Fricatives and stops or affricates have more negative pooled speech retention than vowels (Table 3). Their noise retention is only slightly more negative. The extra speech attenuation is therefore visible even though all four classes undergo substantial noise suppression. Figure 1 shows how the contrast varies with input SNR in cafeteria noise; means approach zero as the mixture becomes cleaner.

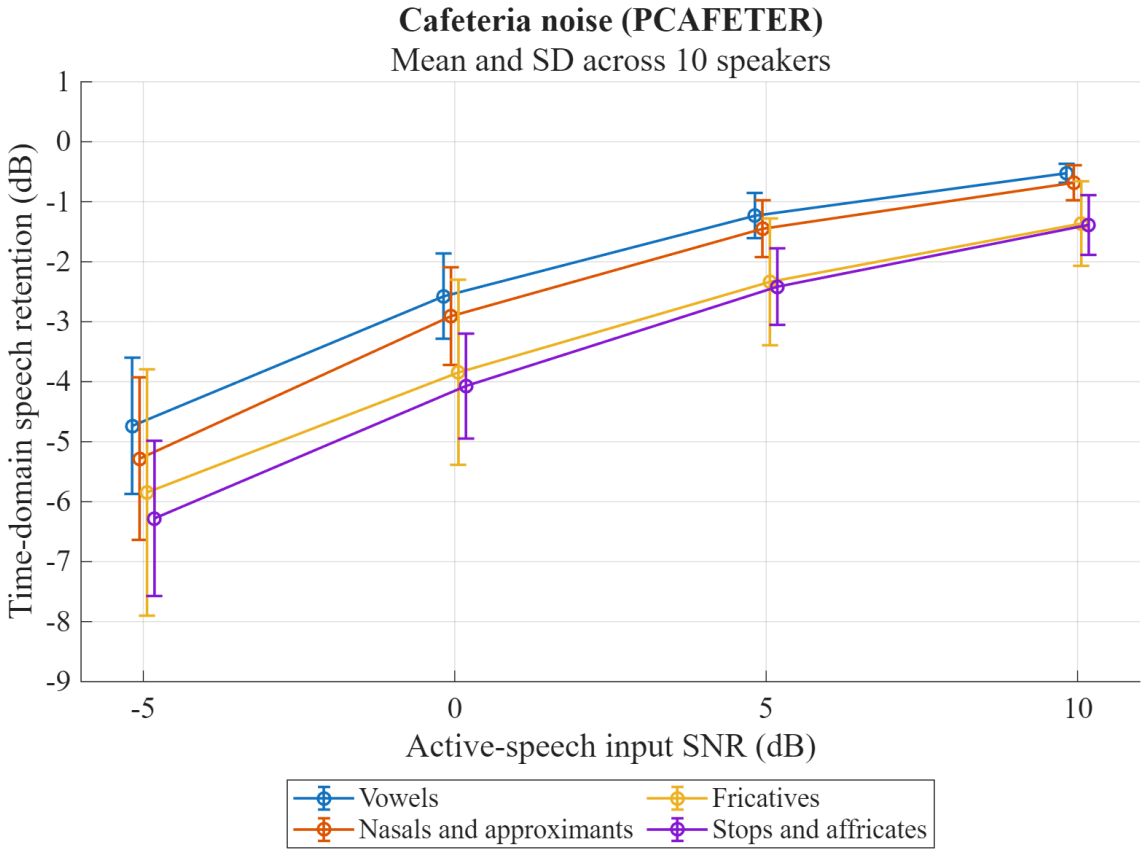


Figure 1. Time-domain speech retention in cafeteria noise. Bars show SD across ten speaker means. Horizontal offsets separate the class markers; all classes use the same four input SNRs.

At −5 dB in cafeteria noise, retention is -4.74 dB for vowels, -5.85 dB for fricatives and -6.28 dB for stops and affricates. Relative to vowels, paired changes are −1.11 dB (95% interval [−2.08, −0.25]) and −1.54 dB ([−2.29, −0.80]), respectively.

Mean segmental error-SNR change is positive in every speech-class/noise/SNR condition. At −5 dB, noise-averaged gains are 8.17 dB for vowels, 8.66 dB for nasals and approximants, 9.50 dB for fricatives and 9.57 dB for stops and affricates. Larger gains coexist with greater speech attenuation.

4.3 Gain and spectral preservation

Across all conditions, fricatives have an unweighted mean gain of 0.465 versus 0.450 for vowels (Table 3). Energy weighting reverses that comparison: 0.749 versus 0.875. Uniform averaging includes bins carrying little clean speech energy. At 0 dB input SNR, averaging noise conditions within speaker gives a mean silence gain of 0.248, below vowels (0.395) and fricatives (0.411). This provides a quantitative non-speech control.

Pooled mean gains below 1 kHz, at 1–4 kHz and at 4–8 kHz are 0.536, 0.446 and 0.461 for fricatives, versus 0.615, 0.496 and 0.374 for vowels. The ranking reverses in the highest band, so a single average gain conceals frequency-dependent differences.

Pooled mixture LSD is 6.45 dB for vowels and 6.82 dB for fricatives, while speech-component LSD is 4.43 and 3.74 dB respectively. Fricatives are therefore not worst on every distortion measure. Frame-based 4–8 kHz retention is −6.08 dB for fricatives, −6.91 dB for vowels, −8.06 dB for nasals and approximants and −7.55 dB for stops and affricates. This compares classes within one band; it does not establish whether fricatives lose more high-frequency than low-frequency energy.

Figure 2 aligns the gain with phonetic classes. Figures 3 and 4 show the vowel and fricative with the most assigned frames in this utterance. This selection improves spectral averaging but does not represent the shortest consonants. Each clean spectral peak sets a common reference for all three curves.

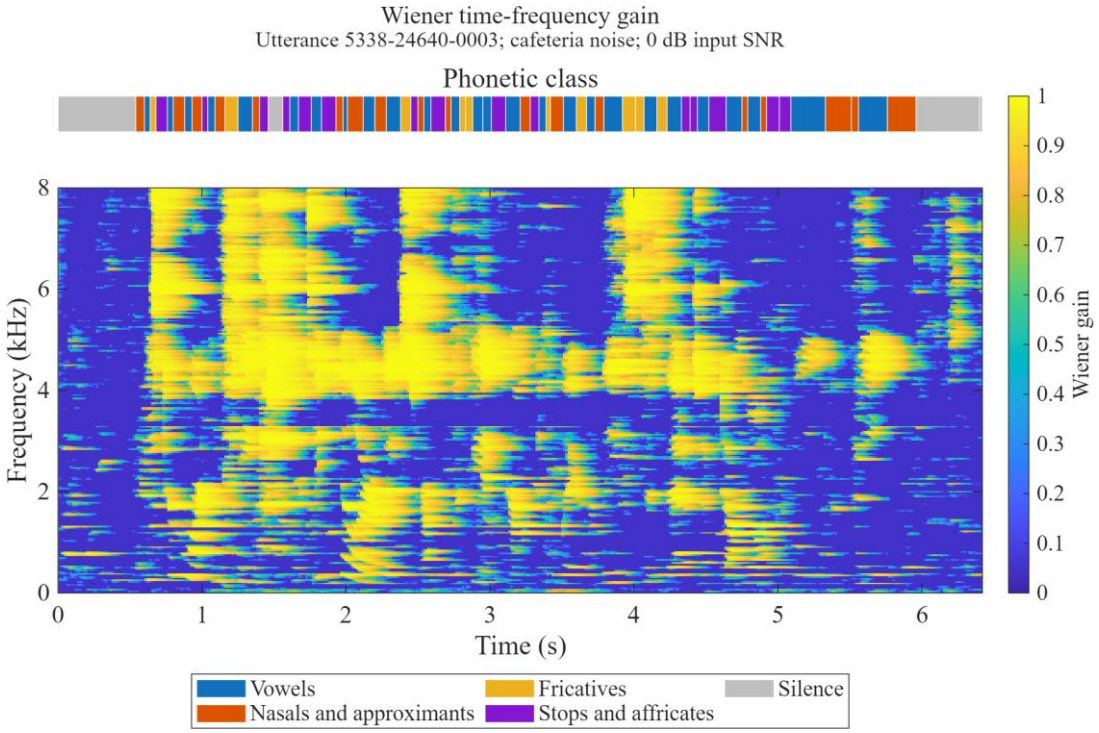


Figure 2. Wiener gain for utterance 5338-24640-0003 in cafeteria noise at 0 dB. The phonetic-class strip shares the time axis of the 0–8 kHz gain map.

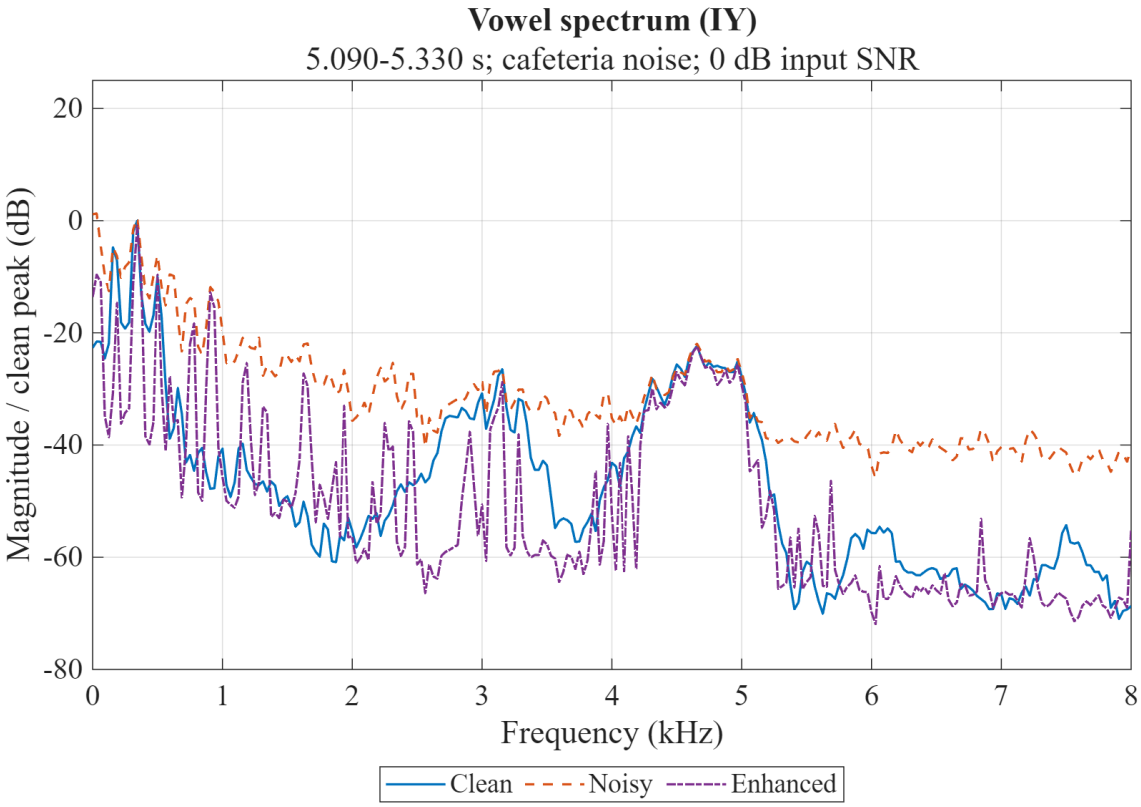


Figure 3. Clean, noisy and enhanced spectra for the vowel labelled IY at 5.09–5.33 s. Curves are RMS magnitudes over the assigned frames, expressed relative to the clean spectral maximum.

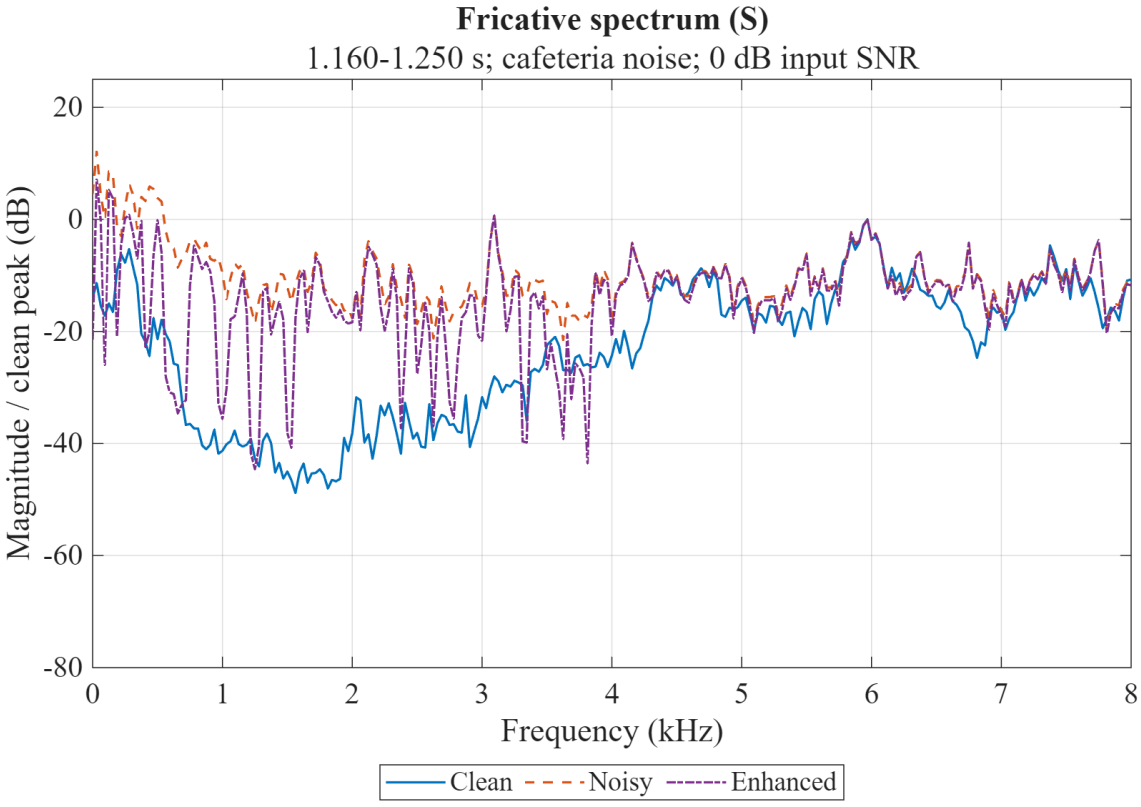


Figure 4. Corresponding spectra for the fricative labelled S at 1.16–1.25 s. High-frequency structure is comparatively well retained in this example, while the enhanced spectrum still differs from the clean reference at lower frequencies.

4.4 Window length comparison

Table 4. Paired changes from 32 ms to 16 ms, pooled within speaker. Intervals are exploratory 95% speaker-bootstrap intervals.

Measurement	Mean change	95% interval
Global error-SNR improvement (dB)	+0.063	[-0.043, +0.166]
STOI	-0.0093	[-0.0110, -0.0077]
Global noise retention (dB)	-0.362	[-0.491, -0.249]
Vowel speech retention (dB)	-0.477	[-0.561, -0.396]
Nasal/approximant speech retention (dB)	-0.609	[-0.768, -0.474]
Fricative speech retention (dB)	-0.593	[-0.711, -0.486]
Stop/affricate speech retention (dB)	-0.628	[-0.767, -0.484]

Shortening the window reduces the class-average fraction of boundary-crossing frames from approximately 42–46% to 19–21%. It nevertheless lowers speech retention in every class and lowers STOI (Table 4). Global error-SNR improvement changes by only +0.063 dB, with an interval containing zero. More noise is suppressed, but more speech is attenuated as well. The 32 ms configuration is retained as the baseline; the result does not establish that it is optimal for other data or settings.

5 Discussion

5.1 Interpretation of the main findings

The tested Wiener rule preserves broad phonetic classes unequally in energy terms. The contrast is clearest between vowels and the fricative or stop/affricate groups under difficult noise. At the same time, global and segmental error SNR can improve. This answers the signal-level part of the research question: a favourable overall score can accompany selective attenuation of the known desired component. The experiment has not established how much of that attenuation changes phoneme or word perception.

The enhanced error combines speech modification with residual noise. Its squared norm contains the energy of both terms and their cross-term. It can fall because interference is removed even while speech retention becomes more negative. The global energy sum also weights strong speech regions more heavily, whereas the class analysis first averages phone scores. Consequently, their numerical values and rankings describe different aspects of performance.

The energy-weighted gain helps explain why an unweighted average can be misleading. Fricatives can receive high gain in many weak bins while their energetic bins are attenuated. However, their pooled 4–8 kHz retention is less negative than that of the other classes. This band-specific ranking does not measure how attenuation varies across frequencies within fricatives. Individual spectra illustrate local behaviour; they do not establish a mechanism for every token.

5.2 Relation to previous work

The separation of utterance and phone-level outcomes follows the motivation of Monir and colleagues [3]. Their findings also include benefits for some sibilants, which is compatible with the absence of a universal fricative disadvantage here. The numerical results cannot be compared directly because the systems, acoustic conditions and metrics differ. This study contributes a

controlled single-channel example and an explicit same-gain component analysis rather than a replication of their multichannel ranking.

Kim and Loizou [2] caution against equating attenuation with an intelligibility penalty. The present energy losses and small mean STOI decrease therefore remain separate observations; their coexistence does not establish causation. The spectral work of Kong and colleagues [4] and Shadle and colleagues [5] supports examining frequency structure, but a broad band-energy ratio cannot measure preservation of every cue relevant to fricative identity. The 8 kHz upper limit further restricts that inference.

5.3 Window length and signal preservation

The shorter-window experiment addresses the inclusion of adjacent phones within a frame. It reduces boundary crossing but increases speech attenuation and lowers STOI. The experiment cannot isolate the effect of reduced boundary mixing: the shorter window also reduces true frequency resolution and changes the sequence of power observations, even with a matched smoothing time constant. The small error-SNR gain therefore provides insufficient reason to replace the 32 ms baseline when speech preservation is the priority.

5.4 Limitations and further work

Ten speakers with one utterance each form a small convenience sample. Equal speaker weights prevent token counts from dominating the summary, but do not separate class from duration, speaking style, original energy or phonetic context. The derivative alignments are structurally validated rather than hand checked. Broad categories combine different sounds, and stop intervals do not isolate bursts. Bootstrap intervals describe this sample and remain exploratory, especially across many condition-specific contrasts.

Only one recording represents each noise environment, and the prefix noise estimate is fixed. Temporal or spectral mismatch may influence performance, but the experiment does not measure stationarity or attribute the cafeteria result to one noise property. The gain floor and power smoothing are not exhaustively tuned. LSD also depends on its magnitude floor, while energy ratios are unstable for nearly silent segments; explicit thresholds reduce that problem without eliminating sensitivity to the measurement definition.

A useful next step is to add several utterances per speaker and inspect a stratified sample of short consonant boundaries. A controlled comparison of noise estimation or gain floor could then use a separate development set, with the same paired speech/noise measurements on held-out material. Listening-based phoneme or word scores would be required to test whether the observed signal changes affect recognition. These additions address the present uncertainties without changing the research question.

6 Conclusion

Across 240 controlled mixtures, the baseline improved mean whole-utterance error SNR by 6.18 dB while leaving mean STOI slightly lower. Same-gain decomposition revealed greater pooled desired-speech attenuation for fricatives and stops or affricates than for vowels. Neither unweighted gain nor high-frequency retention reproduced that ranking, showing why component energy, frequency structure and error-based scores must be interpreted together. The 16 ms experiment reduced boundary mixing but worsened speech retention and STOI, so 32 ms remains the baseline. The study answers the question of unequal signal preservation for the selected data; perceptual and population-wide conclusions remain open.

Reproducibility

The implementation was tested with MATLAB R2025b and Signal Processing Toolbox; Audio Toolbox supplies STOI. The project includes the selected audio, labels and manifests. Running main rebuilds the baseline; validateDesign checks the signal path and metrics; compareWindows runs both window configurations and their paired comparison. The project-site and repository links appear on the first page. Result tables retain individual conditions, speaker means and calculation settings.

References

[1] P. C. Loizou, Speech Enhancement: Theory and Practice, 2nd ed. CRC Press, 2013. doi: 10.1201/b14529

[2] G. Kim and P. C. Loizou, "Gain-induced speech distortions and the absence of intelligibility benefit with existing noise-reduction algorithms," Journal of the Acoustical Society of America, vol. 130, no. 3, pp. 1581–1596, 2011. doi: 10.1121/1.3619790

[3] N.-E. Monir, P. Magron and R. Serizel, "A phoneme-scale assessment of multichannel speech enhancement algorithms," Trends in Hearing, vol. 28, 2024. doi: 10.1177/23312165241292205

[4] Y.-Y. Kong, A. Mullangi and K. Kokkinakis, "Classification of fricative consonants for speech enhancement in hearing devices," PLOS ONE, vol. 9, no. 4, e95001, 2014. doi: 10.1371/journal.pone.0095001

[5] C. H. Shadle, W.-R. Chen, L. L. Koenig and J. L. Preston, "Refining and extending measures for fricative spectra, with special attention to the high-frequency range," Journal of the Acoustical Society of America, vol. 154, no. 3, pp. 1932–1944, 2023. doi: 10.1121/10.0021075

[6] C. H. Taal, R. C. Hendriks, R. Heusdens and J. Jensen, "An algorithm for intelligibility prediction of time-frequency weighted noisy speech," IEEE Transactions on Audio, Speech, and Language Processing, vol. 19, no. 7, pp. 2125–2136, 2011. doi: 10.1109/TASL.2011.2114881

[7] V. Panayotov, G. Chen, D. Povey and S. Khudanpur, "LibriSpeech: An ASR corpus based on public domain audio books," ICASSP, pp. 5206–5210, 2015. doi: 10.1109/ICASSP.2015.7178964

[8] J. Thiemann, N. Ito and E. Vincent, "DEMAND: A collection of multi-channel recordings of acoustic noise in diverse environments," version 1.0, 2013. doi: 10.5281/zenodo.1227121

[9] ChangelingLab, "LibriSpeech Segment," Hugging Face Datasets, accessed 1 November 2026. https://huggingface.co/datasets/changelinglab/librispeech-segment

[10] MathWorks, "stft" and "istft," Signal Processing Toolbox documentation. https://www.mathworks.com/help/signal/ref/stft.html and istft